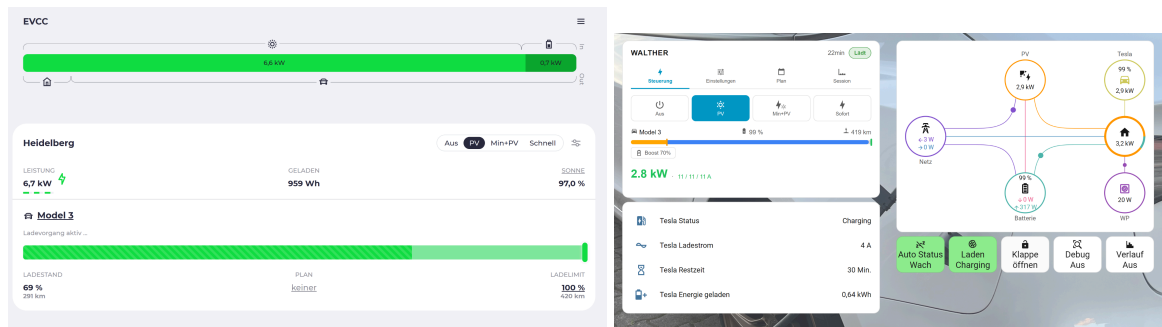


Charge-on-solar for Tesla:

EVCC with Home Assistant and Bluetooth via ESP32 (without cloud)



|                                   |           |
|-----------------------------------|-----------|
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## 1. Installing the ESP32:

There are separate instructions for

this: [https://docs.google.com/document/d/1W33jPQH4uAfSzb8rHvg7sBFm9zY\\_D-149qrtlnRN2xQ/edit?usp=sharing](https://docs.google.com/document/d/1W33jPQH4uAfSzb8rHvg7sBFm9zY_D-149qrtlnRN2xQ/edit?usp=sharing)

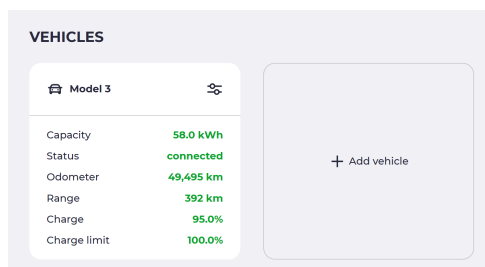
Attention: In Chapter 4: In the configuration, the # characters must be removed from the package evcc.yml:

```
packages:
  remote_package_files:
    url: https://github.com/PedroKTFC/esphome-tesla-ble/
    files:
      - packages/base.yml
      - packages/common.yml
      - packages/project.yml
      - packages/client.yml
      # - packages/listener.yml # Uncomment this to scan find your VIN BLE MAC address
      - packages/evcc.yml # Uncomment this if you use the ESP for EVCC charge control
```

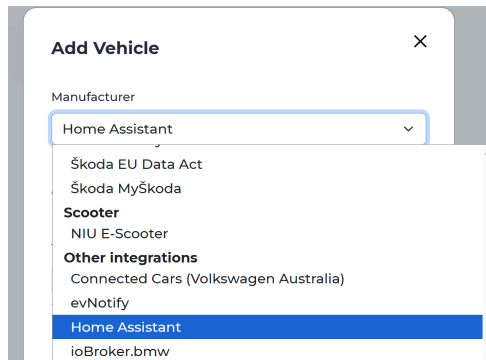
## 2. Connect EVCC

### 2.1. Setting up the car

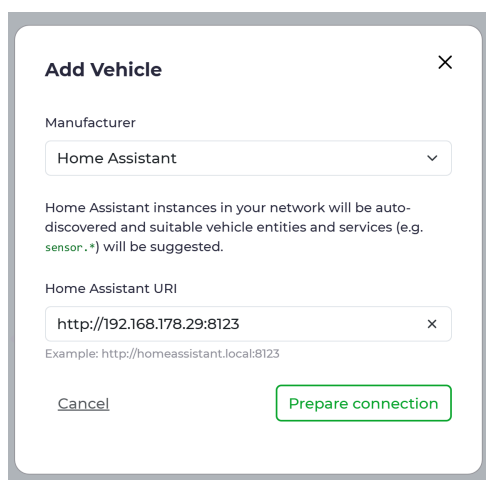
In EVCC, click on More/Configuration, then on “Add vehicle”:



We then select “Home Assistant” as the manufacturer (far down):



Next, the Home Assistant server is specified. It is usually found automatically; otherwise, the server's IP address is entered here.



Now click “Prepare connection”. The first time you do this, a Home Assistant login window will appear, where you need to enter a valid HA username and password. Don't worry, the username is not stored in EVCC, only a “persistent login token”.

From here on, sensors need to be entered. Vehicle name and battery capacity are entered manually. I've allocated 2 kWh of degradation from the nominal capacity. The nominal value is also quite accurate.

Edit Vehicle



Manufacturer

Home Assistant

Home Assistant instances in your network will be auto-discovered and suitable vehicle entities and services (e.g. `sensor.ble_f549c4_charge_level`) will be suggested.

Title

Model 3

Will be displayed in the user interface

Icon

 [change](#)

Will be displayed in the user interface

Home Assistant URI

http://192.168.178.29:8123

Example: http://homeassistant.local:8123

State of charge [%]

sensor.tesla\_ble\_f549c4\_charge\_level

Example: sensor.vehicle\_soc

Entity ID for the vehicle's battery state of charge in percent

Battery capacity optional

58 kWh

Example: 50

Remaining range [km] optional

sensor.tesla\_ble\_f549c4\_range

Example: sensor.vehicle\_range

Entity ID for the vehicle's remaining range in kilometers

Charging status optional

sensor.tesla\_ble\_f549c4\_charging\_state

Example: sensor.vehicle\_charging

Entity ID for charging status (A=disconnected, B=connected, C=charging)

Target state of charge [%] optional

number.tesla\_ble\_f549c4\_charging\_limit

Example: number.vehicle.target\_state\_of\_charge

Entity ID for the vehicle's target state of charge in percent (`number` or `input_number` entity)

Odometer [km] optional

sensor.tesla\_ble\_f549c4\_odometer

Example: sensor.vehicle\_odometer

Entity ID for the vehicle's odometer reading in kilometers

Climatisation active optional

binary\_sensor.tesla\_ble\_f549c4\_climate

Example: binary\_sensor.vehicle\_climate

Entity ID for the vehicle's climatisation state (`binary_sensor` with on/off state)

Finish time optional

sensor.tesla\_ble\_f549c4\_minutes\_to\_limit

Example: sensor.vehicle\_finish\_time

Entity ID for the estimated charging finish time (ISO8601 or Unix timestamp)

Service to start charging optional

switch.tesla\_ble\_f549c4\_charger

Example: script.vehicle\_start\_charge; switch.vehicle\_charger

Entity ID for a script or a switch that starts charging the vehicle. Only useful with the [docs.evcc.io](#). If a switch is provided, Service to stop charging must be left empty.

Service to stop charging optional

Example: script.vehicle\_stop\_charge

Entity ID for a script that stops charging the vehicle. Only useful with the [docs.evcc.io](#)

Service to wake up vehicle optional

Example: script.vehicle\_wakeup; button.wakeup\_vehicle

Entity ID for a script or a button that wakes up the vehicle

Set charging current [A] optional

number.tesla\_ble\_f549c4\_charging\_amps

x

Example: number.vehicle\_charging\_current

Entity ID for setting the maximum charging current in amperes (number or input\_number entity)

Hide advanced settings ▲

Maximum number of phases optional

---

▼

Used to determine the minimum solar surplus and plan duration.

Default charging mode optional

Solar

▼

Applied when a vehicle is connected. Leave empty to keep the current mode.

Minimum current optional

1

A

Example: 6

The minimum current per connected phase that can be used

Maximum current optional

16

A

Example: 16

The maximum current per connected phase that can be used

Maximum charging power hint optional

11000

W

Example: 10000

Defines the maximum charging power of the vehicle. Helps improve charge plan accuracy when the vehicle typically uses less than the offered current or supports higher single-phase current compared to three-phase. The offered current of the loadpoint is not affected.

RFID identification optional

↻

Used to assign the vehicle to a loadpoint via RFID card. The identifier is shown on the loadpoint. Copy the value from there. [Learn more.](#)

Priority optional

Priority of the loadpoint or vehicle in relation to other loadpoints or vehicles for allocating surplus energy (higher values, higher priority).

1A current control optional

no

yes

Vehicle supports 1A current steps only

Supports streaming optional

no

yes

Streaming data is received asynchronously

Charge on connection optional

no

yes

Charger will enable charging for short time when vehicle is connected, irrespective of configured charge mode. This is useful for vehicles that require power supply when connecting.

Status: successful

validate

|              |           |
|--------------|-----------|
| Capacity     | 58.0 kWh  |
| Status       | connected |
| Odometer     | 49,495 km |
| Range        | 392 km    |
| Charge       | 95.0%     |
| Charge limit | 100.0%    |

Delete

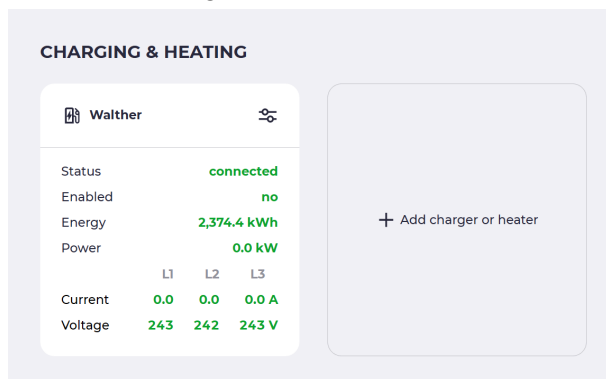
Save

The number of phases and the current must be specified manually. The absolute minimum for Tesla is 1A, but for efficiency reasons I use 2A.

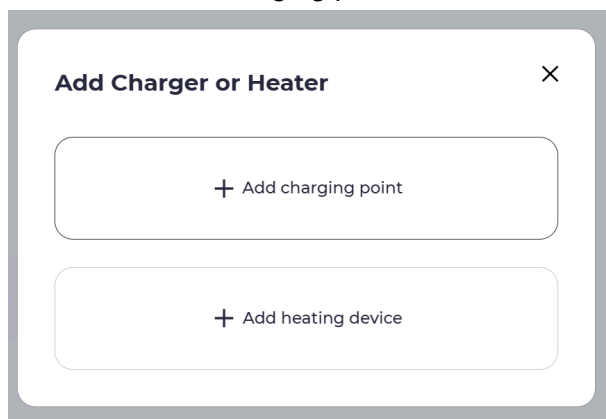
Finally, click “validate”. Plausible values must be displayed when the car is within range. Then, click “Save”.

## 2.2. Setting up the wallbox:

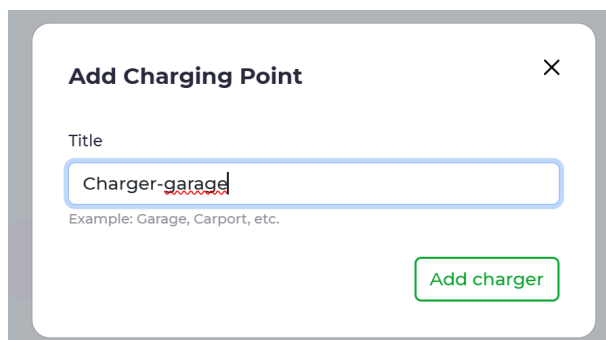
Click “Add charger or heater”:



Then click “Add charging point”:



Assign a name (e.g., manufacturer name, garage left, carport,...) and select “Add charger”



Select “Home Assistant Charger” again (this time near the top) and click “Prepare connection”:

Add Charger

×

Manufacturer

Home Assistant Charger

Home Assistant instances in your network will be auto-discovered and suitable charger entities and services (e.g. `sensor.*`, `switch.*`, `number.*`) will be suggested.

Home Assistant URI

http://192.168.178.29:8123

Example: `http://homeassistant.local:8123`

Cancel

Prepare connection

The entities are again selected via autocomplete:

Edit Charger

ⓘ ×

Manufacturer

Home Assistant Charger

Home Assistant instances in your network will be auto-discovered and suitable charger entities and services (e.g. `sensor.*`, `switch.*`, `number.*`) will be suggested.

Home Assistant URI

http://192.168.178.29:8123

Example: `http://homeassistant.local:8123`

Charging status sensor

sensor.tesla\_ble\_f549c4\_wb\_ladestatus\_iso2

Example: `sensor.charger_status`  
Entity ID for charging status (A=ready, B=connected, C=charging)

Enabled status sensor

switch.tesla\_ble\_f549c4\_charger\_evcc

Example: `binary_sensor.charger_enabled`  
Entity ID for enabled state (`sensor`, `binary_sensor` or `switch` with on/off or true/false state)

Enable switch

switch.tesla\_ble\_f549c4\_charger\_evcc

Example: `switch.charger_enable`  
Entity ID for enable/disable control (`switch` or `input_boolean`)

Maximum current entity [A]

number.tesla\_ble\_f549c4\_charging\_amps\_evcc

Example: `number.charger_max_current`  
Entity ID for setting maximum current in amperes (`number` or `input_number` entity)

Moving on – the power output is intentionally left blank here! The car provides a value, but it's rounded to the nearest kW, which is very inaccurate. EVCC can calculate the power itself from voltage and current. Notice: The current and voltage sensors are entered thrice on 3-phase-systems. On single- or dual-phase systems, leave the unused sensors empty.

Hide advanced settings ▲

Power entity optional

▼

Example: sensor.charger\_power  
Entity ID for instantaneous power measurement in watts

Energy entity optional

sensor.tesla\_ble\_f549c4\_charge\_energy\_added

×

Example: sensor.charger\_energy  
Entity ID for cumulative energy measurement in kWh

L1 current entity optional

sensor.tesla\_ble\_f549c4\_charge\_current

×

Example: sensor.charger\_current\_l1  
Entity ID for L1 current measurement in amperes

L2 current entity optional

sensor.tesla\_ble\_f549c4\_charge\_current

×

Example: sensor.charger\_current\_l2  
Entity ID for L2 current measurement in amperes

L3 current entity optional

sensor.tesla\_ble\_f549c4\_charge\_current

×

Example: sensor.charger\_current\_l3  
Entity ID for L3 current measurement in amperes

L1 voltage entity optional

sensor.tesla\_ble\_f549c4\_charge\_voltage

×

Example: sensor.charger\_voltage\_l1  
Entity ID for L1 voltage measurement in volts

L2 voltage entity optional

sensor.tesla\_ble\_f549c4\_charge\_current

×

Example: sensor.charger\_voltage\_l2  
Entity ID for L2 voltage measurement in volts

L3 voltage entity optional

sensor.tesla\_ble\_f549c4\_charge\_current

×

The last entities:



L1 voltage entity optional

sensor.tesla\_ble\_f549c4\_charge\_voltage x

Example: sensor.charger\_voltage\_l1  
Entity ID for L1 voltage measurement in volts

L2 voltage entity optional

sensor.tesla\_ble\_f549c4\_charge\_voltage x

Example: sensor.charger\_voltage\_l2  
Entity ID for L2 voltage measurement in volts

L3 voltage entity optional

sensor.tesla\_ble\_f549c4\_charge\_voltage x

Example: sensor.charger\_voltage\_l3  
Entity ID for L3 voltage measurement in volts

Phase switching entity optional

Example: select.charger\_phases  
Entity ID for 1p/3p phase switching (select entity with options "1" and "3")

Heating device optional

no yes

Shows °C instead of %

Integrated device optional

no yes

Integrated device. No charging sessions

Status: unknown validate

Cancel Validate & save

L3 Spannungsentität optional

sensor.tesla\_ble\_f549c4\_charge\_voltage x

Beispiel: sensor.charger\_voltage\_l3  
Entitäts-ID für L3 Spannungsmessung in Volt

Phasenumschaltungs-Entität optional

Beispiel: select.charger\_phases  
Entitäts-ID für 1p/3p Phasenumschaltung (Select-Entität mit Optionen "1" und "3")

Wärmeerzeuger optional

nein ja

Zeigt °C anstatt % an

Integriertes Gerät optional

nein ja

Fest angeschlossenes Gerät. Keine Ladevorgänge

Status: unbekannt prüfen

Löschen Überprüfen & speichern

After clicking "Check", values should reappear. It's normal for very low voltage values to appear; expect 2-6V when you are not charging, and normal 115 or 230V when charging.

Status: successful

validate

|         |                          |                  |                    |
|---------|--------------------------|------------------|--------------------|
| Status  | <span>connected</span>   |                  |                    |
| Enabled | <span>no</span>          |                  |                    |
| Energy  | <span>2,374.4 kWh</span> |                  |                    |
| Power   | <span>0.0 kW</span>      |                  |                    |
|         | L1                       | L2               | L3                 |
| Current | <span>0.0</span>         | <span>0.0</span> | <span>0.0 A</span> |
| Voltage | <span>242</span>         | <span>242</span> | <span>242 V</span> |

Delete

Save

Click “Save”.

### 3. PV system configuration

EVCC can query most PV systems directly; alternatively, the data can also be retrieved via Home Assistant. For my Huawei system, there's a very comprehensive integration that queries the system via Modbus/TCP every 15 seconds. Some might use a Modbus proxy, which involves a lot of effort and querying the data a second time. It's simpler to just let Home Assistant do the work and retrieve the data from there. Regarding energy data, it's important to note that EVCC always expects positive values when power or energy flows towards the vehicle, e.g., from the roof to the home network, from the home battery to the home network, or from the grid to the home network. The opposite direction (feed-in or charging the home battery) must be negative. If the integration provides the values in reverse, add a hyphen ("-") before the sensor name; this reverses the measurement direction for EVCC.

#### 3.1 Network meter

Edit Grid Meter

Manufacturer

Home Assistant Meter

Home Assistant instances in your network will be auto-discovered and suitable entities (e.g. `sensor.*`) will be suggested.

Home Assistant URI

http://192.168.178.29:8123

Example: http://homeassistant.local:8123

Power Entity

-sensor.power\_meter\_active\_power

Example: sensor.house\_power  
Entity ID for instantaneous power measurement in watts. The entity must provide numeric values only (e.g., "1234", not "1234 W").

Show advanced settings

Status: unknown

validate

Delete

Validate & save

The grid meter for my Huawei system is a case in point: grid feed-in is positive, grid consumption is negative. That's why there's a "-" before the sensor name. Note: I've seen the opposite meter direction with Huawei devices using Emma. In that case, the "-" is omitted.

### 3.2 Inverter power meter:

Edit Solar Meter

Title

Huawei

Manufacturer

Home Assistant

Home Assistant instances in your network will be auto-discovered and suitable entities (e.g. `sensor.*`) will be suggested.

Home Assistant URI

http://192.168.178.29:8123

Example: http://homeassistant.local:8123

Power Entity

sensor.inverter\_input\_power

Example: sensor.house\_power  
Entity ID for instantaneous power measurement in watts. The entity must provide numeric values only (e.g., "1234", not "1234 W").

Show advanced settings

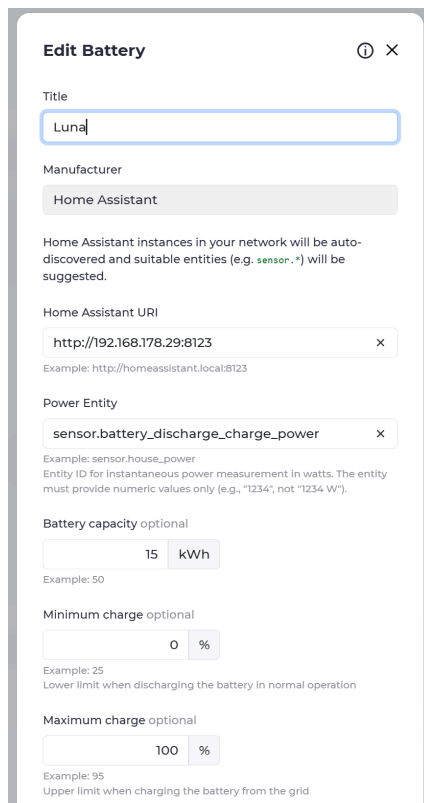
Status: unknown

validate

Delete

Validate & save

### 3.3 Home battery:



The screenshot shows the 'Edit Battery' configuration page in Home Assistant. The page has a title bar with an information icon and a close button. The configuration fields are as follows:

- Title:** A text input field containing 'Luna'.
- Manufacturer:** A dropdown menu showing 'Home Assistant'.
- Home Assistant URI:** A text input field containing 'http://192.168.178.29:8123'. Below it, an example is provided: 'Example: http://homeassistant.local:8123'.
- Power Entity:** A text input field containing 'sensor.battery\_discharge\_charge\_power'. Below it, an example is provided: 'Example: sensor.house\_power'. A note states: 'Entity ID for instantaneous power measurement in watts. The entity must provide numeric values only (e.g., "1234", not "1234 W").'.
- Battery capacity optional:** A numeric input field set to '15' and a unit dropdown set to 'kWh'. Below it, an example is provided: 'Example: 50'.
- Minimum charge optional:** A numeric input field set to '0' and a unit dropdown set to '%'. Below it, an example is provided: 'Example: 25'. A note states: 'Lower limit when discharging the battery in normal operation'.
- Maximum charge optional:** A numeric input field set to '100' and a unit dropdown set to '%'. Below it, an example is provided: 'Example: 95'. A note states: 'Upper limit when charging the battery from the grid'.

## 4. Finish!

Once the PV system is registered, EVCC is functional; nothing more is needed. Optionally, you can also set up the EVCC integration from Marq24, which allows you to display EVCC data, for example, in dashboards.

<https://github.com/marq24/ha-evcc>

And there are some nice maps for Home Assistant from mkshb:

<https://github.com/mkshb/hass-evcc-card>